# A Dual-Stream Coding Fusion Network for Landslide Mapping Based on Multisource Remote Sensing Images

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***Abstract*—The accurate acquisition of fine-scale landslide information is critical for disaster early warning and prevention. However, existing landslide detection methods mainly focus on the spatial characteristics of multisource remote sensing data, while often overlooking the cross-channel semantic dependencies inherent in their fusion. This limitation hinders the comprehensive characterization of landslides in complex environments, leading to insufficient recognition accuracy across different shapes and scales, and resulting in missed detections and false alarms. To address this challenge, we propose a dual-stream coding fusion landslide detection network (DSCF-LDNet), which simultaneously models spatial features and cross-channel semantic dependencies in multisource landslide remote sensing data, thereby enabling the high-precision automatic landslide detection. Specifically, the spatial branch employs a multiscale residual network (MSRNet) to extract fine-grained spatial features, whereas the sequential branch integrates an MSRNet with a long short-term memory module (MSRNet-LSTM) to model cross-channel correlations among multisource features. Subsequently, a feature fusion module is designed to integrate both representations, generating more discriminative landslide features. In the decoding stage, an attention-gated decoder module (AGDM) is designed to enhance multiscale feature interactions, thereby improving the model's sensitivity to landslide targets. Experiments on the self-constructed Bailong River Basin dataset demonstrate that DSCF-LDNet achieves a precision of 80.84%, a recall of 77.89%, an *F*1-score of 79.35%, and a mean intersection over union (mIoU) of 81.74%, surpassing benchmark models such as SCDUNet++, STUNet, and Attention-UNet. Tests on Landslide4Sense and Lanzhou Urban Area datasets further confirm the model's robustness across diverse geological settings. Overall, DSCF-LDNet effectively learns both spatial features and interchannel semantic dependencies from multisource remote sensing data, enabling accurate landslide mapping for diverse shapes and scales, and provides a practical framework for large-scale landslide monitoring.**

***Index Terms*—Attention-gated decoder module (AGDM), landslide mapping, long short-term memory (LSTM), multiscale residual network (MSRNet), multisource remote sensing data.**

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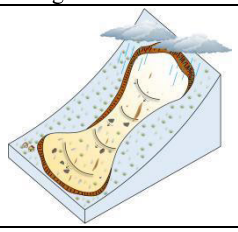
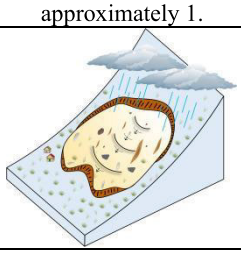
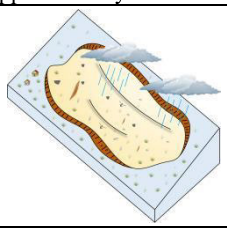
## I. Introduction

With the intensification of global climate change, extreme weather events such as heavy rainfall have become frequent, significantly increasing the risk of landslides and the associated loss of life and property [1], [2]. Landslides exhibit diverse morphological characteristics and are commonly classified from two perspectives. Based on the geometric shape, landslides can be classified into elongated, equidistant, and wide-spreading types [3]. In terms of scale, they are categorized as large, medium, and small events [4]. A summary of these classification criteria is provided in Table I. High-precision historical landslide cataloging is essential not only for issuing early warnings and minimizing casualties during disaster events but also for providing critical information on landslide locations and spatial extent [5]. This information can support governmental agencies and relevant departments in guiding the construction of protective infrastructure and the development of emergency response strategies. Consequently, one of the primary objectives of current landslide research is to develop high-precision identification methods that can rapidly and accurately acquire landslide. These advancements enhance landslide identification efficiency and response capabilities, thereby mitigating potential social and economic losses [6], [7].

Landslide identification methods are generally categorized into three types: traditional manual field surveys, semiautomatic recognition based on machine learning (ML), and automatic recognition based on deep learning (DL) [8]. Traditional methods rely on manual field investigations to update landslide inventory maps. While this approach yields

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TABLE I
LANDSLIDES ARE CLASSIFIED ACCORDING TO SHAPE AND SCALE

| | Elongated | Equidistant | Wide-spreading |
|---|---|---|---|
| | The aspect ratio $R$ of the landslide is greater than 1. | The aspect ratio $R$ of the landslide is approximately 1. | The aspect ratio $R$ of the landslide is approximately less than 1. |
| Landslide shape |  |  |  |
| Landslide scale | Large ($m^3$) | Medium ($m^3$) | Small ($m^3$) |
| | $100\times10^4 \le V < 1000\times10^4$ | $10\times10^4 \le V < 100\times10^4$ | $V < 10\times10^4$ |

relatively accurate results, it is labor-intensive and time-consuming, making it inefficient for large-scale or rapid identification [9]. These limitations are further exacerbated in complex environments, hindering the timely acquisition of landslide. Semiautomatic landslide identification using ML techniques has been widely adopted due to its ability to learn the diverse characteristics of landslides. Common methods include multivariate statistical analysis [10], decision tree [11], support vector machine (SVM) [12], and artificial neural network (ANN) [13]. However, these approaches often require extensive manual feature engineering and parameter tuning. Moreover, in mountainous regions where landslides vary significantly in spectral, geometric, and textural features, and where complex backgrounds exist, misclassifications are frequent [14], [15].

In recent years, with the widespread application of DL techniques in remote sensing image analysis, especially the rapid development of convolutional neural networks (CNNs), significant progress has been made in landslide identification tasks. Owing to their advantages in feature extraction accuracy and automation, CNN-based approaches have effectively overcome the limitations of traditional landslide recognition methods [13], [16]. These methods can extract high-level semantic features from imagery, eliminating the dependence on handcrafted feature design in conventional ML and greatly enhancing image classification and semantic segmentation performance. In addition, the construction of large-scale and high-resolution remote sensing imagery datasets, along with the development of adaptive multiple field-of-view fusion networks for remote sensing image semantic segmentation, has provided a solid data and model foundation for landslide detection in complex geomorphic environments [17], [18]. Meanwhile, the development of end-to-end remote sensing image semantic segmentation frameworks, such as REST [19] and PSPNet [20], has further facilitated the automated recognition of fine-grained targets in remote sensing imagery.

Building upon this, researchers have designed a variety of CNN-based network architectures for landslide detection and delineation, achieving superior performance in terms of both accuracy and generalization. Lei et al. [21] proposed a fully convolutional network (FCN) approach for landslide cataloging mapping, incorporating pyramid pooling to enhance spatial feature extraction. Huang et al. [22] employed the DeepLabv3+ semantic segmentation algorithm to identify the thaw slump of the Beibailu River on the Tibetan Plateau. Qi et al. [23] proposed an ResU-Net model that integrates residual learning, applying it to landslide mapping in Tianshui, Gansu, China. Building on the spectral and terrain data, Wu et al. [24] constructed a landslide recognition model that combines the strengths of CNN and Transformer architectures. Xie et al. [25] proposed a novel landslide extraction framework that incorporates context-related features, specifically designed for aerial image datasets. Wang et al. [26] introduced a hybrid YOLO + U-Net-based method for landslide recognition, which significantly enhances detection performance under complex background conditions. Yang et al. [27] proposed a new landslide semantic segmentation model named ResU-SENet to address the urgent needs of large-scale co-seismic landslide mapping. Dong et al. [28] proposed a progressive landslide identification model (MFFLnet) for rapid co-seismic landslide detection.

In summary, landslides can be identified more accurately using DL methods based on optical remote sensing imagery. However, several challenges remain to be addressed, primarily include the following.

1) Landslides exhibit a wide range of geometric shapes (such as elongated, equidistant, and wide-spreading) and varying spatial scales (large, medium, and small). Optical remote sensing imagery alone often fails to effectively capture the heterogeneous characteristics of landslides, particularly in complex geographical environments, thereby the limiting detection accuracy. For instance, terrain features, such as roads and bare soil, which share similar spectral characteristics with landslides, are prone to misclassification [29]. In addition, surfaces with intricate textural patterns may produce circular boundary shapes that resemble landslide signatures [30]. The inherent variability in landslide shapes and scales further complicates the classification process, resulting in misidentification and reducing extraction accuracy.
2) Most existing DL-based landslide recognition models primarily focus on the spatial characteristics of

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multisource remote sensing data, while overlooking the cross-channel feature dependencies that arise from multisource data fusion. This oversight hinders the characterization of landslide features in complex environments, leading to decreased detection accuracy for landslides of varying shapes and scales, as well as increased rates of missed and false detections. Moreover, during the encoding process of DL models, the progressive deepening of network layers typically reduces the receptive field, making it difficult to preserve fine-grained features of landslide regions, particularly for long, narrow, or small-scale landslides, thus increasing false negative (FN) and false positive (FP) rates.

To address the challenges of landslide identification in complex geological environments, this study selects Bailong River Basin as the experimental scene, constructs a dedicated landslide dataset, and proposes a novel landslide detection framework, the dual-stream coding fusion landslide detection network (DSCF-LDNet). The main contributions of this study are as follows.

1) We established an open and high-precision remote sensing landslide dataset for the Bailong River Basin, comprising 773 annotated landslide samples and 16 multisource remote sensing images. Geological hazard experts employed a "dual-annotation and cross-validation" strategy for sample labeling and verification, further validating the results through field surveys and historical data comparison with ensure the accuracy and reliability of the annotations.
2) A fine-grained landslide detection framework DSCF-LDNet, based on the dual-stream encoding of spatial and sequential features, is proposed. By incorporating a dual-stream encoding module and an attention-gated decoder module (AGDM) decoding module, the model significantly enhanced its ability to represent the landslide features under complex terrain conditions.
3) A comprehensive validation was conducted on the self-developed Bailongjiang landslide dataset, as well as the publicly available Landslide4Sense [31] and Lanzhou urban landslide datasets [32]. The results demonstrate that the proposed method achieves the superior performance with strong robustness and generalization capabilities.

## II. Study Area and Data

### A. Study Area

The Bailong River Basin is located at the intersection of the Qinghai-Tibet Plateau, the Loess Plateau, and the Sichuan Basin, featuring a unique geographical setting and complex geological structure (see Fig. 1). The region lies within the central-northern segment of China's north–south seismic belt, with dense fault structures and frequent seismic activity. It is characterized by intense valley incision, steep terrain, and concentrated heavy rainfall, making it one of the most geohazard-prone areas in China. Landslides in this region are widespread and recurrent, influenced by multiple complex and interacting controlling factors [33]. They mainly occur in mid- to low-elevation valley zones, with types including bedrock landslides, accumulation landslides, river terrace loess landslides, and landslides developed in tertiary and Cretaceous laterite deposits. These landslides exhibit diverse morphologies (elongated, equidistant, and wide spreading) and cover a range of scales from large to small. The region's complex topography, dense vegetation, and variable meteorological conditions pose significant challenges for accurate landslide identification and monitoring. Therefore, the Bailong River Basin was selected as the study area for model training and evaluation, providing an ideal testing ground for validating the applicability and extraction accuracy of the model under real-world conditions.

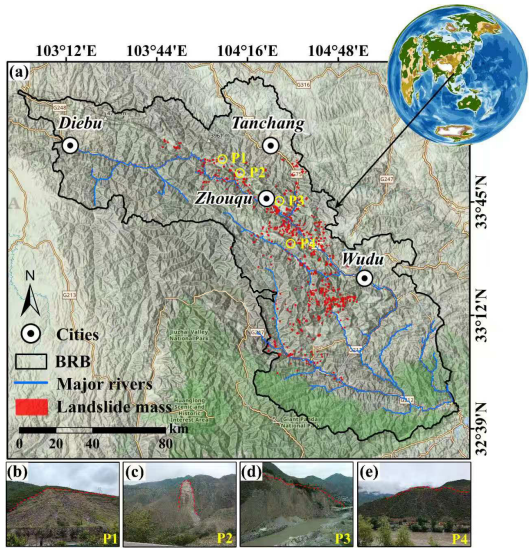



Fig. 1. Overview map of the study area. (a) Schematic map showing the geographical scope of the Bailong River Basin and the distribution of landslide samples. (b)–(e) Photographs from field investigations showing typical landslide sites.

### B. Data

*1) Landslide Data:* Landslide vector boundaries were delineated through visual interpretation of high-resolution Google Earth imagery, supplemented by regional historical landslide records and field investigations for validation [34]. In this study, a total of 773 historical landslide polygons were manually annotated by experts through interpretation of Google Earth imagery. For landslide detection tasks, these vector annotations are typically converted into binary mask images, where pixels labeled as 1 denote landslide regions and pixels labeled as 0 denote nonlandslide areas. These binary masks serve as ground-truth labels for supervised learning and model evaluation.

*2) Sentinel-2A:* Sentinel-2A, an optical remote sensing satellite operated under European Space Agency's (ESA) Copernicus program, provides high-resolution multispectral imagery with spatial resolutions ranging from 10 to 60 m [35]. With a 10-day revisit cycle and a swath width of 295 km, Sentinel-2A enables frequent and large-scale Earth surface monitoring. The satellite is equipped with 13 spectral bands

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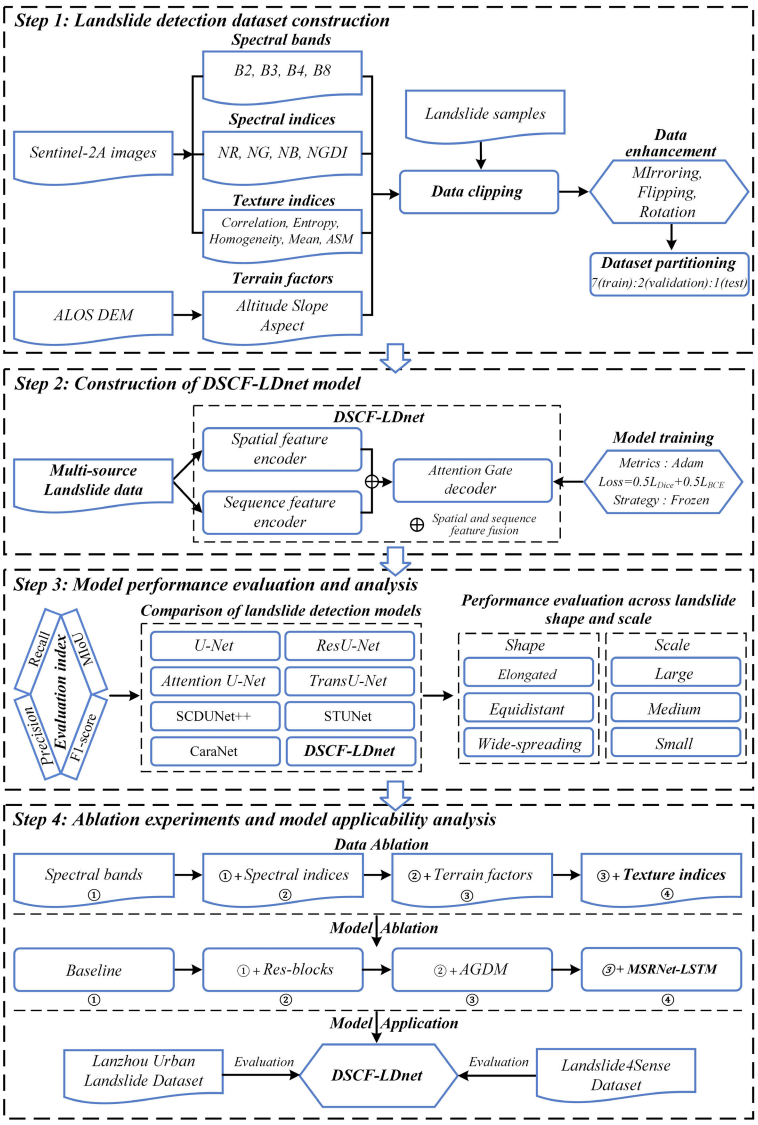



Fig. 2. Flowchart of this research framework. The workflow consists of four steps: 1) construction of the landslide detection dataset; 2) construction of the DSCF-LDNet model; 3) evaluation and analysis of model performance; and 4) ablation studies and analysis of model applicability.

spanning the visible-to-near-infrared (NIR) regions, offering comprehensive data support for applications such as change detection and landslide mapping. Following an unprecedented heavy rainfall event in the Bailong River Basin in August 2020, new landslide deformations were triggered, while many pre-existing landslides experienced accelerated movement and substantial damage. Cloud-free Sentinel-2A imagery acquired on April 29, 2021, was selected for analysis in this study.

*3) Digital Elevation Model (DEM):* DEM data were obtained from 12.5-m resolution terrain products generated by the Phased-Array L-Band Synthetic Aperture Radar (PALSAR) onboard the Advanced Land Observing Satellite (ALOS) of Japan. These high-precision DEM data effectively capture topographic characteristics relevant to landslide analysis, including altitude, slope, and aspect, thereby enabling more accurate delineation of landslide boundaries. In this study, slope and aspect features are derived from DEM were integrated into a landslide detection framework to enhance the spatial characterization.

## III. Research Methods

The research framework of this article is shown in Fig. 2, composed of four primary steps.

1) Calculation of multidimensional landslide feature.
2) Construction of a dual-stream coding landslide recognition model.
3) Model performance evaluation and analysis.
4) Ablation experiments and model application.

### A. Calculation of Multidimensional Feature Data for Landslides

*1) Optical Band Selection:* The optical remote sensing data utilized in this study are cloud-free Sentinel-2A data acquired on April 29, 2021 and provided by ESA. The Sentinel-2A sensor includes 13 optical bands, of which bands B2–B4 and B8 offer a spatial resolution of 10 m. To improve the accuracy of landslide boundary extraction, these four bands were selected as the primary sources of spectral data.

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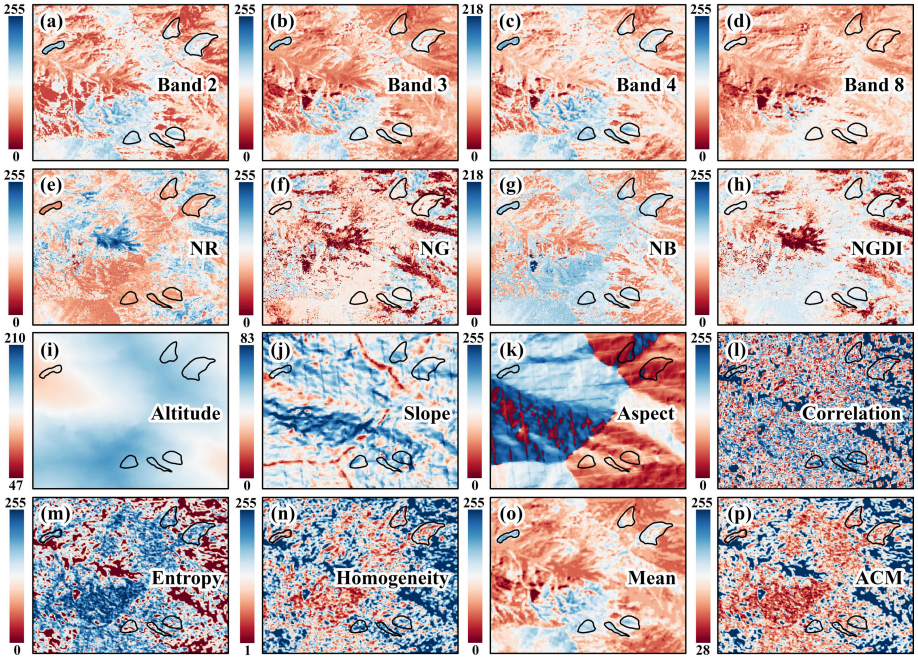



Fig. 3. Visualization of multidimensional features used in this study. (a)–(d) Optical bands. (e)–(h) Spectral indices. (i)–(k) Terrain factors. (l)–(p) Texture indices.

Following data preprocessing, the selected bands were resampled and cropped based on the experimental scene, subsequently exported in the PNG format for use in the model input. Representative images of these bands for a typical area are shown in Fig. 3(a)–(d).

*2) Spectral Index Calculation:* The spectral index variation analysis provides an effective means for the identification and extraction of landslide boundaries in optical imagery. Following a landslide, phenomena such as vegetation degradation, soil exposure, and altered moisture content produce distinctive spectral signatures in affected areas. These variations can be quantified through spectral indices. In this study, the selected indices include the normalized blue band (NB), normalized red band (NR), normalized green band (NG), and the normalized green–red difference index (NGRDI) [36]. The equations used to calculate these indices that are listed in Table II. After calculating each spectral index, the results were resampled to a 10-m spatial resolution and exported in PNG format for subsequent analysis [see Fig. 3(e)–(h)].

*3) Terrain Factor Calculation:* Terrain factors play a fundamental role in landslide initiation and spatial distribution, and serve as important auxiliary variables for landslide mapping. In this study, altitude, slope, and aspect were extracted from the ALOS DEM data with a native resolution of 12.5 m. The DEM was resampled to align with the target resolution of 10 m, and both slope and aspect were derived using standard geomorphometric techniques. The resulting terrain feature layers were exported in the PNG format [see Fig. 3(i)–(k)].

*4) Texture Index Calculation:* Landslide-induced mass movement and surface deformation can cause significant changes in image texture. For instance, the movement and mixing of soil, rock, and debris during a landslide can introduce distinct texture patterns in areas that previously exhibited homogeneous surface characteristics. In this study, five texture indices were selected, including correlation, entropy, homogeneity, mean, and angular second moment (ASM) [36], as summarized in Table II. These texture indices were derived from Sentinel-2A imagery using the gray level co-occurrence matrix (GLCM) method implemented in ENVI 5.6. The texture images were resampled to the target resolution of 10 m and exported as PNG format. The resulting texture feature layers are shown in Fig. 3(i) and (m)–(p).

### *B. Dual-Stream Encoding Landslide Recognition Model*

In complex geological and geomorphological environments, landslides exhibit highly diverse characteristics, posing significant challenges for accurate identification. Most existing approaches primarily emphasize the spatial features of multisource landslide data, while overlooking the interchannel semantic dependencies that emerge during multisource data fusion. This oversight limits the comprehensive characterization of landslides in complex terrain. As a result, the recognition accuracy for landslides of varying shapes and scales is insufficient, as well as the increased rates of missed and false detections. Motivated by the prior research is employing two-stream architectures for landslide mapping [37]. This study proposes a novel DSCF-LDNet, with its architecture illustrated in Fig. 4. The overall framework is based on the U-Net architecture. To effectively capture both spatial detail features and cross-channel semantic relationships from multisource landslide data, the encoder is designed with two parallel branches. The first branch incorporates residual modules to extract local spatial structure, while the second integrates an MSRNet-LSTM module to model contextual dependencies across feature channels, thereby enhancing the

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TABLE II
DETAILED INFORMATION ON LANDSLIDE IDENTIFICATION IMAGE DATA

| Data types | Element | Formula | Formula declare | Effect |
|---|---|---|---|---|
| Optical bands | Band 2 (10 m, Fig. 3(a)) | / | / | These bands provide more detailed landslide characteristics and boundary information. |
| | Band 3 (10 m, Fig. 3(b)) | / | / | |
| | Band 4 (10 m, Fig. 3(c)) | / | / | |
| | Band 8 (10 m, Fig. 3(d)) | / | / | |
| Spectral index | Normalized red band (NR, Fig. 3(e)) | $NR = \frac{R}{B+G+R}$ | R、G and B refer to the red, green, and blue bands of the remote sensing imagery | The spectral index will change before and after landslides |
| | Normalized green band (NG, Fig. 3(f)) | $NG = \frac{G}{B+G+R}$ | | |
| | Normalized Blue Band (NB, Fig. 3(g)) | $NB = \frac{B}{B+G+R}$ | | |
| | Normalized green-red difference index (NGDI, Fig. 3(h)) | $NGDI = \frac{G \bullet R}{G+R}$ | | |
| Terrain factor | Altitude, Fig. 3(i) | / | Where, $Z$ denotes the elevation value, $x$ represents horizontal distance, $y$ represents vertical distance | Terrain factors have important effects on the formation and distribution of landslides |
| | Slope, Fig. 3(j) | $Slope = \arctan\left(\sqrt{\left(\frac{\partial Z}{\partial x}\right)^2 + \left(\frac{\partial Z}{\partial y}\right)^2}\right)$ | | |
| | Aspect, Fig. 3(k) | $Aspect = \arctan\left(\frac{\frac{\partial Z}{\partial y}}{\frac{\partial Z}{\partial x}}\right)$ | | |
| Texture index | Correlation, Fig. 3(l) | $Correlation = \sum_{i,j}^{N-1} P_{i,j}\left[\frac{(i-\mu_i)(i-\mu_i)}{\sqrt{(\sigma_i^2)(\sigma_j^2)}}\right]$ | Here, $i$ and $j$ represent the row and column indices of the GLCM, the $P_{i,j}$ represents the probability of gray levels $i$ and $j$ appearing together in neighboring pixels within a specified window of the original image. | The texture index of the landslide location is different from that of the non-landslide area |
| | Entropy, Fig. 3(m) | $Entropy = \sum_{i,j=0}^{N-1} P_{i,j}\left(-\ln P_{i,j}\right)$ | | |
| | Homogeneity, Fig. 3(n) | $Homogeneity = \sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1+(i-j)^2}$ | | |
| | Mean, Fig. 3(o) | / | | |
| | Angular Second Moment (ASM, Fig. 3(p)) | $ASM = \sum_{i,j=0}^{N-1} \left(P_{i,j}\right)^2$ | | |

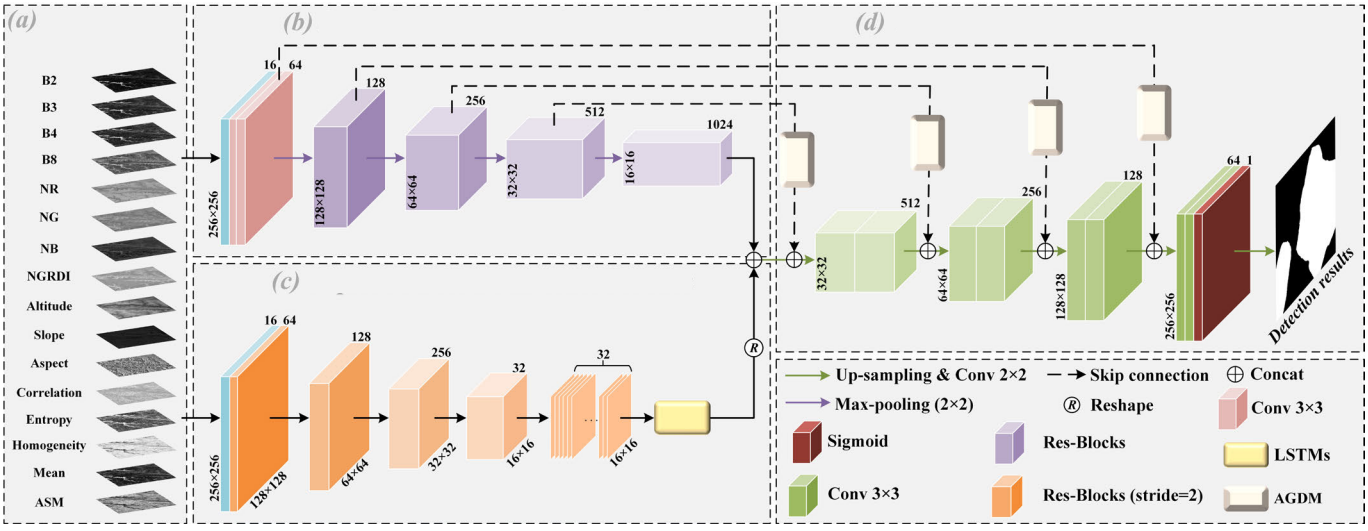



Fig. 4. Architecture of the proposed DSCF-LDNet model. (a) Multisource landslide data input. (b) Spatial feature encoder module. (c) Sequence feature encoder module. (d) Landslide feature decoder module.

representational capacity of the fused features beyond what is achievable with residual modules alone. A fusion module is then employed to combine spatial and sequential features, generating rich deep feature representations. In the decoder, an AGDM was introduced to enhance the fusion between deep semantic features from the decoder and low-level spatial detail features from the encoder, thereby improving sensitivity to landslide regions. Finally, a $1 \times 1$ convolutional layer followed by a sigmoid classifier is used to distinguish between landslide and nonlandslide areas, yielding the final prediction map. The

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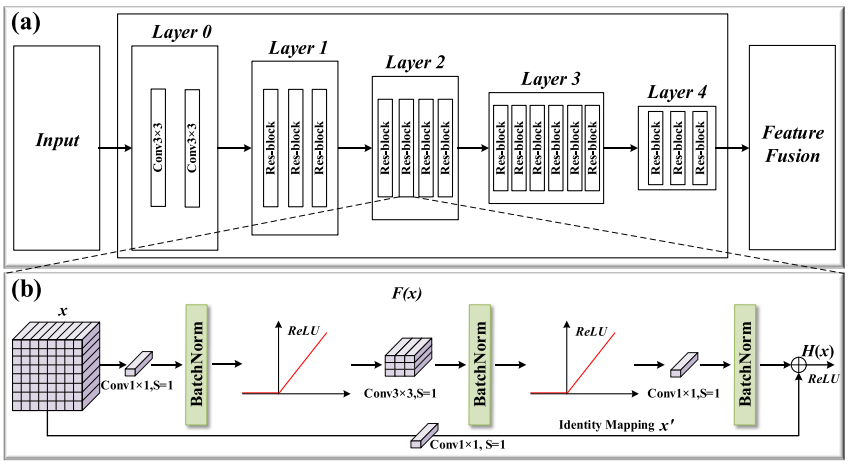



Fig. 5. Spatial feature extraction flowchart. (a) Internal composition of the spatial feature extraction module. (b) Internal composition of res-blocks.

detailed architecture and components of the proposed model are presented in Sections III-B1–III-B3.

*1) Encoder Module:* The encoder in this study adopts a dual-branch architecture, consisting of a spatial feature encoder and a branch for modeling cross-channel dependencies among multisource features. A dedicated fusion module is then designed to effectively integrate the spatial and cross-channel features, thereby generating rich, high-level feature representations.

*a) Spatial dimension feature encoder module:* To effectively capture the inherent spatial correlations present in multisource landslide data, the spatial encoding branch is designed to enhance the ability of the model to interpret and analyze spatial features. Conventional U-Net encoders typically consist of stacked convolutional and pooling layers. However, because of its relatively simple structure, it may suffer from underfitting when processing large or complex datasets, thus limiting its capability to capture the intricate features essential for accurate landslide identification [38]. However, the extraction of detailed image features is crucial for accurate landslide identification. To enhance the integration of contextual and fine-grained information during feature extraction, while alleviating performance degradation due to increased network depth, this study introduces a novel encoder backbone tailored for spatial feature representation.

The proposed encoder is composed of four downsampling steps, as illustrated in Fig. 5(a). Each stage consists of two standard convolutional layers followed by four residual convolutional structure layers designed to extract rich semantic features at multiple levels of abstraction. Initially, the input data are mapped into a high-dimensional feature space via the two standard convolutional layers. These layers employ a kernel size of $3 \times 3$ with a stride of 1, each followed by a batch normalization (BN) layer [39] and a rectified linear unit (ReLU) activation layer. Subsequently, the feature maps are projected into an even higher dimensional space via four residual convolutional structure layers, enabling the extraction of deeper semantic information. The four residual layers comprise 3, 4, 6, and 3 res-blocks, respectively. As illustrated in Fig. 5(b), each res-block consists of two $1 \times 1$ convolutional layers and one $3 \times 3$ convolutional layer, all with a stride of 1. The use of stacked small-kernel convolutions facilitates the effective extraction of fine-grained features associated with small-scale landslides, while simultaneously enriching the contextual semantic information essential for recognizing complex patterns. Furthermore, during the spatial feature extraction process, each downsampling operation doubles the number of feature channels while halving the spatial resolution, thereby enabling the capture of higher level semantic representations.

The detailed architecture of each residual block designed in this study is illustrated in Fig. 5(b). It contains three convolutional layers, each followed by a BN layer. ReLU activation functions were inserted after the first and second BN layers to enhance the nonlinearity of feature transformation and improve the model's representation capability [40]. In addition to the main convolutional pathway, a residual connection is established between the input and output of the block. Along this residual path, a $1 \times 1$ convolution is applied to perform identity mapping between the input and output feature vectors. The mathematical formulation of the residual block can be expressed as follows:

$$H(x) = F(x) + x' \tag{1}$$

where $x$ is the input feature, $F(x)$ represents the main branch convolutional output, $x'$ represents the residual mapping output, and $H(x)$ is the output of the whole residual block.

*b) Sequence dimension feature encoder module:* The pixel-level cross-channel feature extraction module is designed to learn the interdependencies among various landslide factor channels, enabling the network to capture the relationships between different factors at the pixel level. This module consists of a multiscale residual network (MSRNet) and a long short-term memory (LSTM) network [41]. MSRNet comprises four downsampling stages, each containing a residual convolutional block composed of two $3 \times 3$ convolutional layers. During feature extraction, the spatial dimensions of the feature maps are halved after each residual convolutional block. The LSTM module is composed of three stacked LSTM layers, which are employed to model the cross-channel dependencies among multisource landslide features. The overall architecture of this branch network is illustrated in Fig. 6(a).

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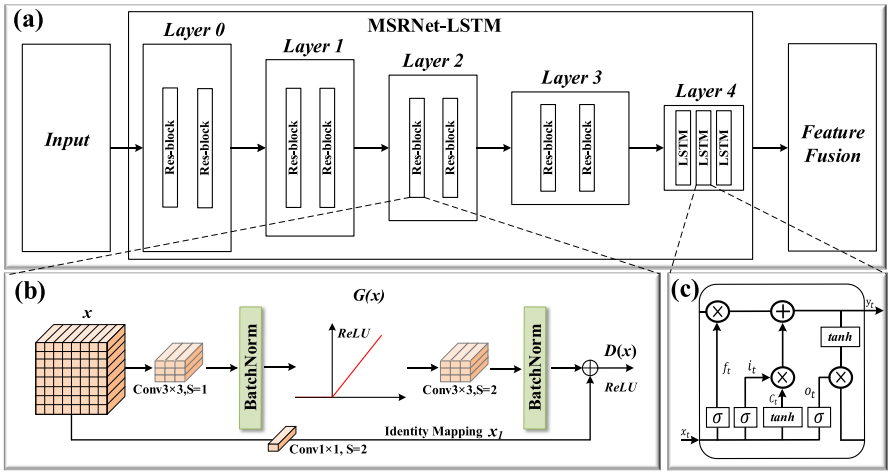



Fig. 6. Sequential feature extraction flowchart. (a) MSRNet-LSTM network composition. (b) Internal composition of res-blocks (stride = 2). (c) LSTM network architecture.

For the acquired 16-channel landslide image dataset, MSRNet is first applied to perform feature extraction and spatial dimensionality reduction, producing multidimensional feature maps. Each feature channel of the output feature map is then flattened into a 1-D feature vector. These vectors are concatenated in channel order to form a pixel-level feature sequence, which matches the input format required by the LSTM network. The subsequent LSTM layers effectively learns the potential nonlinear relationships and composite patterns among feature channels, thereby generating more discriminative fused feature representations and enhancing the ability of the model to distinguish landslide features. The MSRNet architecture consists of four residual learning units at different scales. Each unit contains two res-blocks, with the structure of a res-block as shown in Fig. 6(b). Each res-block comprises two $3 \times 3$ convolutional layers, each followed by a BN layer to stabilize the input distribution across layers. The ReLU activation function is employed to reduce the overfitting during training. In addition, within each residual learning unit, the second res-block uses a convolution operation with a stride of 2 to perform downsampling. The mathematical formulation of the residual block is provided as follows:

$$D(x) = G(x) + x_1 \tag{2}$$

where $x$ denotes the input feature, $G(x)$ represents the main branch convolutional output, $x_1$ corresponds to the residual mapping output, and $D(x)$ is the output of the whole residual block. After passing through MSRNet for feature extraction and spatial dimensionality reduction, the input data are transformed into a multidimensional feature map of size $16 \times 16$. Due to hardware limitations during the experiment, the final output channel number of these multidimensional features is set to 32.

This study constructs a network of three LSTM layers to model cross-channel dependencies among multisource landslide features, thereby obtaining more stable and complementary feature representations. The architecture of the LSTM unit is depicted in Fig. 6(c). In this structure, $x_t$ stands for input, $y_t$ stands for output, $\sigma$ stands for a fully connected layer with activation function, $f_t, i_t, c_t$, and $o_t$ stand for forget gate, input gate, candidate memory, and output gate, respectively, in the LSTM structure [41]. During the experiments, each LSTM layer is configured with 32 neurons, with other hyperparameters remaining unchanged. In addition, during model training, batch size plays a crucial role in determining the optimization efficiency and training speed of the model [42]. After extensive experimental validations, considering both hardware constraints and the scale of the dataset, the batch size for training the LSTM network is set to 8.

*c) Dual-stream feature fusion:* To integrate the spatial and cross-channel features extracted from the two parallel branches, feature fusion was performed from 1-D and 2-D perspectives. This was achieved via vector concatenation, where the feature representations learned by the spatial encoding branch and sequential encoding branch were combined using the "concat" operation to form a unified feature vector. The fused feature vector then serves as the input to the decoder module, which performs upsampling to generate a final prediction. The corresponding formulation is expressed as follows:

$$E_{\text{concat}} = \left[E_{\text{spatial}}, E_{\text{sequence}}\right] \tag{3}$$

where $E_{\text{spatial}}$ and $E_{\text{sequence}}$ represent the features extracted by the spatial and sequential encoding branch network, respectively.

*2) Decoder Module:* The decoder was designed to reconstruct the prediction outputs from the high-dimensional features obtained during the encoding phase. In the proposed model, the deep features generated by the two-stream encoder fusion serve as input to the decoder, which progressively upsamples these features to produce the final predictions. The decoder of the proposed model consists of four hierarchical decoding modules. Each module includes two $3 \times 3$ convolutional layers, an upsampling operation, and a channel concatenation step. Each convolutional layer is followed by a BN layer and an ReLU activation function. During each upsampling stage, upsampling is performed using an Up-conv2 $\times$ 2 layer, which simultaneously enlarges the spatial resolution and halves the channel dimension through a

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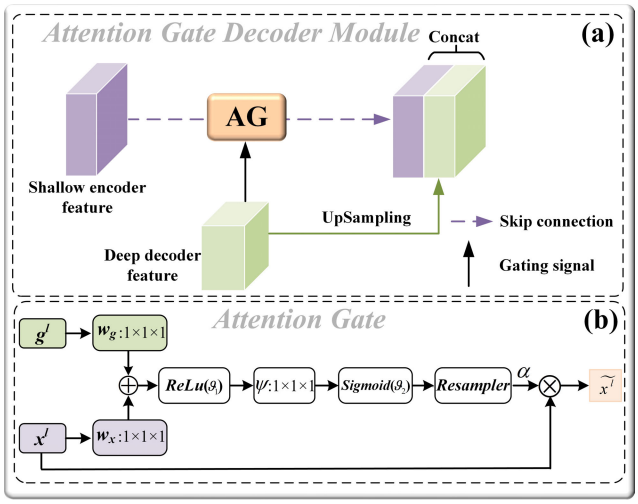



Fig. 7. AG Decoder module. (a) Schematic of the AG decoder module. (b) Detailed internal mechanism of the AG structure.

learnable 2 × 2 transposed convolution. Subsequently, the upsampled feature map is concatenated with the corresponding feature map from the spatial encoder at the same resolution via skip connections, which helps to preserve high-resolution spatial details and improves the final segmentation performance. Finally, a 1 × 1 convolutional layer followed by a sigmoid classifier predicts the pixelwise probabilities of landslide and background, generating segmentation results with the same size as the input image.

In our landslide dataset, since nonlandslide background regions dominate, they introduce substantial noise during training, often resulting in FPs and missed detections of landslide regions, thereby degrading overall recognition performance. To mitigate the interference caused by a nonlandslide background, an attention gate (AG) mechanism is incorporated into the skip connections between the encoder and decoder, forming the AGDM, as illustrated in Fig. 7(a). Specifically, the AG module receives as inputs the deep decoder features and the shallow encoder features from the corresponding resolution level. It computes gating weights to generate a spatial attention map that highlights important features across spatial locations, guiding the model to focus on landslide regions with varying shapes and scales. This attention mechanism suppresses irrelevant background information, strengthens the fusion of shallow encoder and deep decoder features, and ultimately enhances recognition accuracy.

The structure of the AG module is illustrated in Fig. 7(b), where the attention value is computed on a per-pixel basis, and the gate signal determines the regions of interest for the model on a per-pixel basis [43]. To achieve higher accuracy, an additive attention mechanism is employed to calculate attention coefficients, which can be expressed as follows:

$$Q_L = \psi\left(\vartheta_1\left(W_g g^l + W_x x^l + b_g\right)\right) + b_\psi \tag{4}$$

$$\alpha = f_{\text{Res}}\left(\vartheta_2\left(Q_L\right)\right) \tag{5}$$

where $\vartheta_1$ and $\vartheta_2$ are labeled as ReLU and Sigmoid activation functions, respectively. $W_g$ and $W_x$ are the weights, $b_g$ and $b_\psi$ are the bias, and $\psi$ is obtained by applying a 1 × 1 convolutional kernel to the input feature map. $f_{\text{Res}}$ denotes

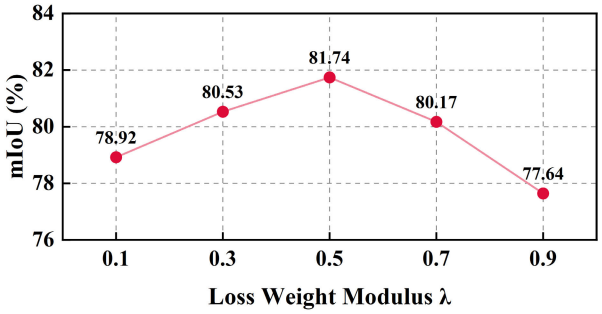



Fig. 8. mIoU performance under different a loss weight $\lambda$.

the resampler operation. The attention coefficient matrix $\alpha$ is obtained through a resampler, and it is multiplied by the input $x^l$ to get the final output. Here, the resampler method used to obtain the attention coefficients is bilinear interpolation. In addition, the AG parameters were updated through back-propagation training.

*3) Loss Function:* In DL model optimization, the loss function quantifies the discrepancy between predicted outputs and ground-truth labels. Its primary role is to guide parameter updates by minimizing the loss value, thereby progressively improving the prediction accuracy. The choice of loss function depends on the specific task. For landslide recognition, binary cross entropy (BCE) loss [23] treats all samples equally and is generally stable. However, in highly imbalanced classification scenarios, such as distinguishing landslide from dominant nonlandslide areas, it tends to bias the model toward the majority background class, leading to neglected landslide regions and degraded overall performance. Dice loss [44] effectively addresses class imbalance by maximizing overlap between predicted and true masks, but it may be difficult to accurately delineate complex boundaries or detect small target regions. Consequently, relying solely on either the BCE or the Dice loss may not fully satisfy the accuracy and small-object detection requirements inherent to landslide recognition.

To overcome these limitations, we propose a weighted loss function $L$ that integrates both BCE and Dice losses within a fully supervised learning framework. The weighted loss function combines the stability of BCE and the advantages of Dice loss in solving sample imbalance, while assigning greater weight to landslide areas to emphasize their importance during training. This strategy mitigates the risk of the landslide class being overwhelmed by the background, thereby enhancing precise segmentation of the landslide regions. Loss function $L$ is defined as follows:

$$L = (1-\lambda)\, L_{\text{Dice}} + \lambda L_{\text{BCE}} \tag{6}$$

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=0}^{n}\left[y_i \log_a p_i + (1-y_i)\log_a (1-p_i)\right] \tag{7}$$

$$L_{\text{Dice}} = 1 - \frac{1}{N}\sum_{i=1}^{N}\frac{2\sum_i^N y_i p_i}{\sum_i^N y_i + \sum_i^N p_i} \tag{8}$$

where $\lambda$ represents the weight coefficient assigned to the BCE loss function. When $\lambda$ is set to 0.1~0.9, different weight coefficients are selected with a step size of 0.2 to conduct

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experiments on the datasets used, and the mean intersection over union (mIoU) index is used for performance evaluation. As shown in Fig. 8, the highest mIoU value is obtained when $\lambda = 0.5$. Accordingly, $\lambda$ was set to 0.5 during model training to balance stability and class imbalance effectively. $\lambda$ denotes the total number of classified categories, $y_i$ denotes the label value of pixel $i$th, and $p_i$ indicates the predicted probability of the pixel $i$th.

### C. Experimental Setup

*1) Datasets Cutting:* A landslide dataset was constructed from the Bailong River Basin. Considering both spatial resolution and computational efficiency, each image patch was cropped to 256 × 256 pixels using a sliding window with a stride of 128 pixels. The optical bands, spectral indices, terrain factors, texture indices, and corresponding landslide labels were cropped into aligned subimages.

To enhance the diversity of training samples, mitigate overfitting, and improve the model's generalization capability, data augmentation was applied to both the input subimages and their labels. Specifically, horizontal and vertical flipping, mirroring, and 90° counterclockwise rotation were performed. This process yielded 3271 labeled training samples. The dataset was then randomly divided into training, validation, and testING subsets with a ratio of 7:2:1, resulting in 2290, 654, and 327 images, respectively. All subsets were strictly nonoverlapping to ensure independent evaluation. It is worth noting that the three subsets were mutually exclusive.

*2) Performance Metrics:* To assess the performance of the proposed model in landslide extraction, commonly used metrics [45], including precision, recall, $F1$-score, and mIoU, were used. The calculation methods are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{10}$$

$$F1\text{-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recll}} \tag{11}$$

$$\text{mIoU} = \frac{1}{2} \times \left( \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP} + \text{FN}} \right) \tag{12}$$

where true positive (TP) denotes the correctly identifying landslide targets, FN represents the landslide targets mistakenly classified as nonlandslide. FP indicates the nonlandslide regions incorrectly predicted as landslides, and true negative (TN) corresponds to correctly classified nonlandslide targets.

In addition, two metrics, model parameter count (Params) and floating-point operations per second (FLOPs), were used to evaluate the spatial and computational efficiency of the model [46], which reflects the space and time complexity of the model, thereby indicating its lightweight design and computational performance. Specifically, Params represent the total number of learnable parameters, indicating model complexity, while FLOPs measure the computational cost by quantifying the number of floating-point operations required. All models were evaluated on a single NVIDIA RTX 3090 GPU to ensure consistency and comparability.

*3) Implementation Details:* The proposed method was implemented using the PyTorch framework and executed on a single NVIDIA 3090 GPU, which has 24-GB memory. During training, the batch size was fixed at 4, with an initial learning rate of $\lambda = 10^{-5}$. The Adam optimizer was employed for adaptive learning rate adjustment. To accelerate convergence and prevent overfitting, a frozen training strategy was adopted, in which the encoder weights were initially frozen for the first 80 epochs, followed by fine-tuning of all network layers. The maximum number of epochs was set to 200, and early stopping was applied based on the validation loss.

## IV. Experimental Results and Analysis

### A. Model Performance Analysis

To comprehensively validate the effectiveness of the proposed DSCF-LDNet, we conducted extensive comparative experiments against several representative SOTA semantic segmentation models, including U-Net [47], ResU-Net [48], Attention-UNet [45], TransUNet [49], DeeplabV3+ [50], STUNet [51], CaraNet [52], and SCDUNet++ [24]. To ensure a fair and reliable comparison, all models were trained from scratch based on open source code, under a unified hardware platform and consistent training settings. The results of the comparative experiments are shown in Table III; DSCF-LDNet achieves the highest overall performance across all evaluation metrics, with a precision of 80.84%, recall of 77.89%, $F1$-score of 79.35%, and mIoU of 81.74%. Compared with classical architectures such as U-Net and ResU-Net, the proposed method improves precision by 5.67% and 2.98%, respectively, and increases mIoU by more than 4%, demonstrating superior capability in extracting and learning landslide features in complex terrain environments. Even when compared with advanced models incorporating attention mechanisms or spatial priors, such as Attention-UNet, STUnet, SCDUNet++, and DSCF-LDNet still shows a clear performance advantage. Notably, in comparison with the high-performing landslide detection model SCDUNet++, DSCF-LDNet yields improvements of 2.2%, 2.63%, 1.98%, and 1.8% in precision, recall, $F1$-score, and mIoU, respectively. These enhancements underscore the strong segmentation capability of the model in accurately delineating between landslide and nonlandslide regions.

Furthermore, we evaluated the computational efficiency of all models. DSCF-LDNet achieves high segmentation accuracy while maintaining relatively low complexity, with only 54.72 million parameters and 102.41 GFLOPs. It outperforms high-complexity models such as Attention-UNet (189.51 GFLOPs), TransUNet (105.34M), and Deeplabv3+ (192.87 GFLOPs) in terms of computational efficiency. Although lightweight models such as U-Net, ResU-Net, and CaraNet have fewer parameters, their lower accuracy and relatively high computational cost limit their practical applicability. Overall, these findings confirm that DSCF-LDNet achieves an excellent balance between accuracy and efficiency, making it a promising tool for high-precision landslide detection tasks in complex geographical environments. For all models, the parameter count (Params) and FLOPs are reported with an input image size of 16 × 256 × 256.

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TABLE III
EVALUATION INDICES OF LANDSLIDE IDENTIFICATION RESULTS OF DIFFERENT MODELS

| Model | Precision(%) | Recall(%) | F1-score(%) | mIoU(%) | Params(M) | FLOPs(G) |
|---|---|---|---|---|---|---|
| U-Net | 75.17 | 72.74 | 73.91 | 74.32 | 30.16 | 147.65 |
| ResU-Net | 77.86 | 74.33 | 76.05 | 77.31 | 43.93 | **84.69** |
| Attention-UNet | 78.54 | 75.71 | 77.09 | 78.36 | **33.18** | 189.51 |
| TransUNet | 71.32 | 74.12 | 72.69 | 70.20 | 105.34 | 93.24 |
| Deeplabv3+ | 73.41 | 74.46 | 73.93 | 75.43 | 68.70 | 192.87 |
| STUNet | 78.93 | 74.82 | 76.82 | 78.04 | 41.64 | 135.47 |
| SCDUNet++ | 79.64 | 75.26 | 77.37 | 79.92 | 63.20 | 118.49 |
| CaraNet | 75.36 | 73.64 | 74.50 | 75.07 | 47.59 | 121.37 |
| Ours | **80.84** | **77.89** | **79.35** | **81.74** | 54.72 | 102.41 |

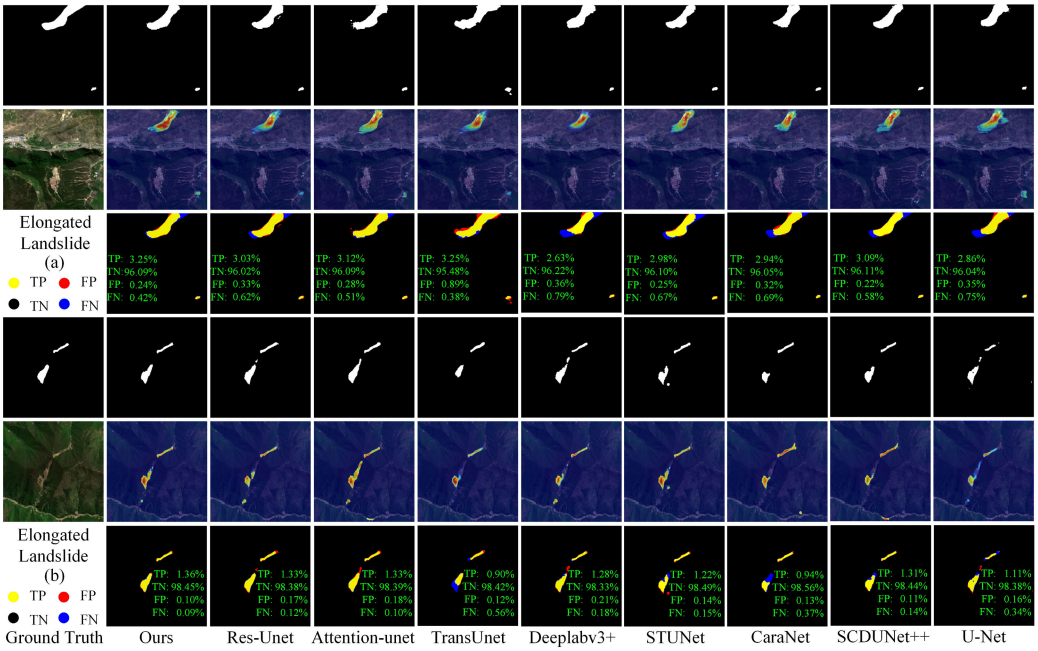



Fig. 9. (a) and (b) Visual comparison of detection results for elongated landslides using different models in two representative scenarios. For each scene, the first row shows the extracted landslide regions. The second row shows the feature visualization heatmaps. The third row shows the corresponding error maps, where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative, respectively.

### *B. Analysis of Landslide Detection Results of Different Shapes*

To comprehensively evaluate the recognition performance and robustness of the proposed model across various landslide morphologies, this study selected two representative scenarios three typical landslide types: elongated, equidistant, and wide-spreading landslides. These scenarios encompass a range of terrain complexity and background interference. A multifaceted comparative analysis was performed using visual interpretations of extraction results, feature visualization heatmaps, and error maps, along with quantitative metrics including TP, FP, TN, and FN, to assess the accuracy of the extraction results for each landslide type and scenario.

For the elongated landslides, Scenario 1 [see Fig. 9(a)] presents a typical instance of an elongated landslide aligned with the terrain direction. This landslide was characterized by a narrow, extended shape amid complex geomorphological surroundings and substantial background noise. Such environments feature steep slope gradients and intricate terrain transitions, which often lead to ambiguous landslide boundaries and increase the difficulty of accurate segmentation. The experimental results demonstrate that the proposed model effectively captures key structural characteristics of the landslide in Scenario 1, particularly along complex and irregular boundaries. It achieves high boundary recognition accuracy and successfully follows the extension trajectory of the landslide, minimizing FPs induced by background clutter. Compared with alternative methods, the proposed model more reliably distinguishes landslide from nonlandslide regions at boundary zones, thereby reducing false identifications. These outcomes highlight the strong robustness of the model to noise and its superior ability for precise boundary extraction.

Scenario 2 [see Fig. 9(b)] involved a landscape with relatively low background interference and simpler terrain conditions. The landslide in this case is well-defined, with sharper transitions between the landslide boundaries and the surrounding terrain. The experimental results indicate that the proposed model performs exceptionally well in this scenario. It accurately delineates the overall landslide shape and

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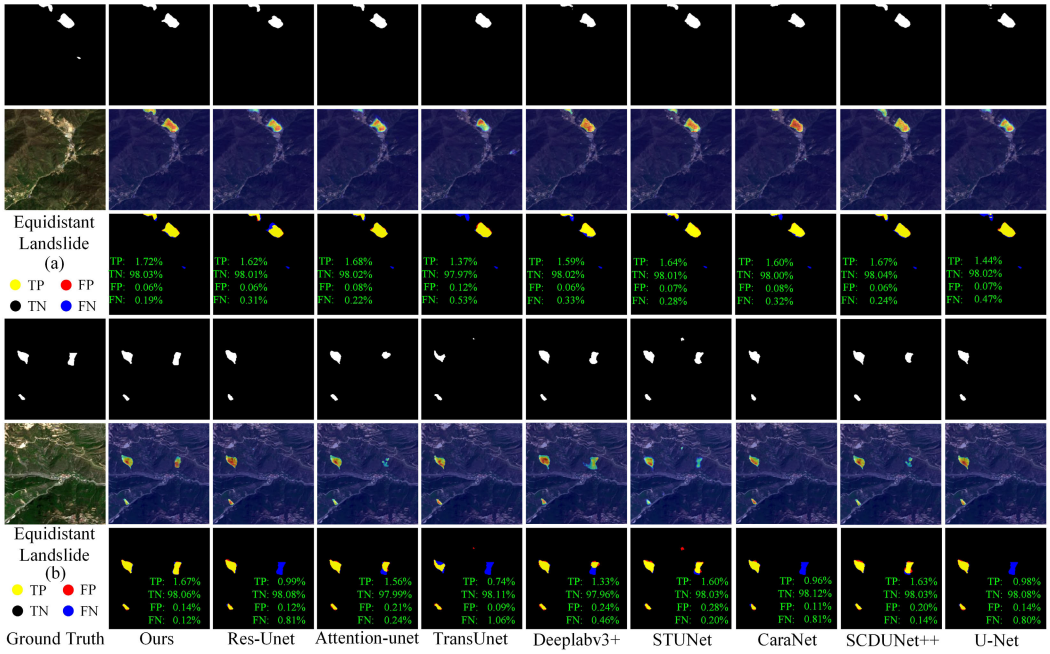



Fig. 10. (a) and (b) Visual comparison of detection results for equidistant landslides using different models in two representative scenarios. For each scene, the first row shows the extracted landslide regions. The second row shows the feature visualization heatmaps. The third row shows the corresponding error maps, where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative, respectively.

effectively captures fine details along the landslide boundaries. Thereby minimizes the misclassification of background noise as part of the landslide. It exhibits a notably reduced FP rate compared with ResU-Net, Attention-UNet, and DeepLabV3+, emphasizing its reliability in relatively straightforward extraction scenarios. In contrast, TransUnet and STUNet tend to underperform in delineating the boundaries of elongated landslides, often relying too heavily on global features. This overreliance results in inaccurate shape extraction or blurred boundary predictions, particularly in areas with notable terrain variability.

Overall, the proposed model consistently exhibits superior feature recognition and boundary delineation capabilities in both complex and simple scenarios involving elongated landslides. It effectively captures critical morphological and structural features even under the challenges of complex topography and background noise, thereby significantly enhancing both extraction accuracy and robustness. These results affirm the strong adaptability and generalization ability of the model in processing elongated landslide types.

For equidistant landslides, Scenario 1 [see Fig. 10(a)] involves a landslide event located in a region characterized by pronounced terrain variability and substantial background noise. Equidistant landslides are typically exhibit characterized by near-symmetrical shapes with relatively consistent width and length, along with well-defined and regular boundaries. However, the presence of complex topography poses challenges to accurate feature extraction. Experimental results show that the proposed model effectively captures the overall morphological characteristics of the landslide in Scenario 1 and accurately delineates the complete boundary. The model demonstrates strong noise resilience, maintains a clear distinction between landslide boundaries, and substantially reduces FPs near the edges. In contrast, models such as TransUNet and Res-Unet exhibit difficulty in handling complex boundary information in this scenario, often leading to the misclassification of background regions as landslides and consequently higher FP rates.

Scenario 2 [see Fig. 10(b)] represents a landslide event in a region with relatively simple terrain and low background interference. The landslide in this setting retains a regular and symmetrical shape with clearly defined boundaries. The experimental results indicate that the proposed model maintains high recognition performance in this scenario as well, accurately extracting the morphology of the landslide and precisely delineating its boundaries. The model excels at suppressing background interference at the landslide–nonlandslide interface and demonstrates strong capability in identifying boundary details, thereby further reducing the FP rate. Compared with other models, the proposed approach exhibits superior boundary recognition and extraction performance for equidistant landslides, particularly in transitional zones where subtle noise or terrain shifts may otherwise degrade the segmentation accuracy.

In summary, the proposed model consistently achieves high accuracy and robustness in recognizing equidistant landslides under varying terrain complexities. It effectively preserves the structural integrity and boundary clarity of landslide regions while significantly minimizing FPs, improving effectiveness, and reliability in equidistant landslide detection tasks.

For wide-spreading landslides, Scenario 1 [see Fig. 11(a)] illustrates a large-scale landslide with an irregular morphology, exhibiting significant lateral expansion and surrounded by complex background features. Due to their extensive spatial distribution and often ambiguous boundaries with adjacent terrain, wide-spreading landslides pose substantial challenges for

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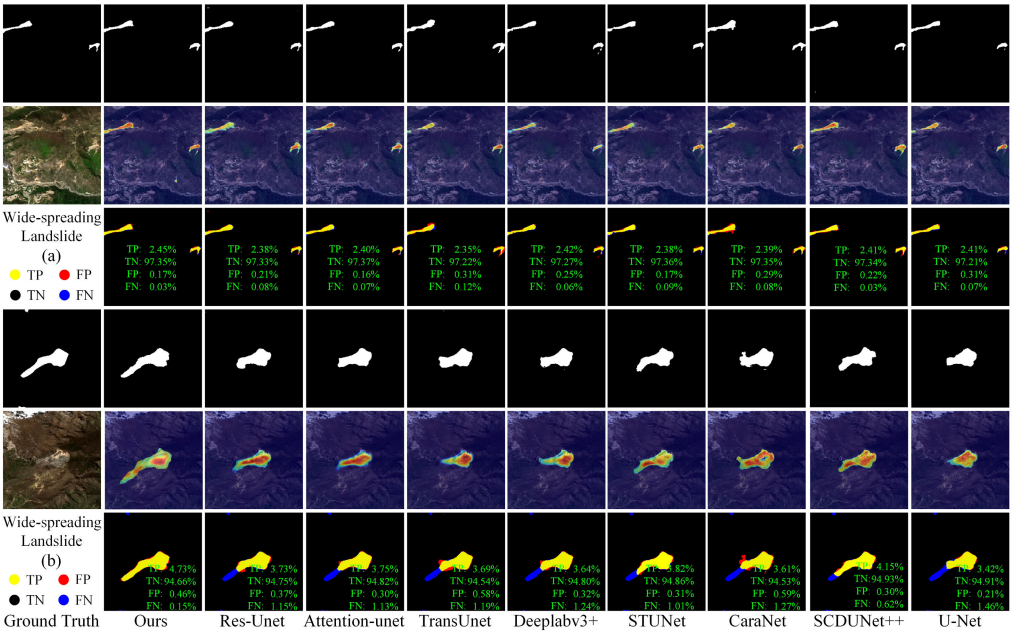



Fig. 11. (a) and (b) Visual comparison of detection results for wide-spreading landslides using different models in two representative scenarios. For each scene, the first row shows the extracted landslide regions. The second row shows the feature visualization heatmaps. The third row shows the corresponding error maps, where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative, respectively.

accurate identification. The experimental results demonstrate that the proposed model effectively captures the dominant morphological features of the landslide in Scenario 1, with particularly strong performance in delineating complex and diffuse boundaries. The model exhibits excellent noise suppression capabilities, significantly reducing FPs caused by the misclassification of adjacent nonlandslide regions. Even in the presence of intricate background textures and variable terrain conditions, the model maintains high precision in boundary extraction, highlighting its adaptability and resilience to noise.

Scenario 2 [see Fig. 11(b)] involves a wide-spreading landslide characterized by more clearly defined contours and distinct boundary transitions. In this setting, the proposed model accurately identifies the global structure of the landslide and effectively captures fine-grained edge details, particularly in areas with complex terrain transitions. The model demonstrates strong discriminative capability in separating landslide from nonlandslide regions, leading to a noticeable reduction in FNs. In comparison, baseline models such as STUNet, SCDUNet++, and U-Net underperform in capturing such boundary nuances, often resulting in incomplete or imprecise delineation and lower overall segmentation accuracy. These models tend to be more vulnerable to interference from terrain variability, whereas the proposed method exhibits greater robustness and consistently minimizes both FN and FP rates, thereby enhancing the detection reliability.

Across both scenarios, the proposed model demonstrates a high level of proficiency in feature extraction and boundary recognition for wide-spreading landslides. It consistently extracts salient landslide features and maintains high segmentation accuracy despite variations in terrain background complexities, with particular advantages in fine boundary delineation.

Overall, the experimental results across landslides of varying geometries, elongated, equidistant, and wide-spreading, affirm the superior accuracy of the proposed model, generalization ability, and robustness. In scenarios with complex topographic features and significant noise, the model effectively suppresses misclassification and captures critical boundary details, greatly improving the landslide recognition reliability. In contrast, traditional models such as U-Net, Deeplabv3+, and STUNet exhibit notable limitations in boundary extraction and noise suppression under similar conditions, resulting in decreased detection performance.

### C. Analysis of Landslide Detection Results of Different Scales

To comprehensively evaluate the effectiveness of the proposed model for extracting landslides of varying spatial scales, this study selected two representative scenarios for each of the large-, medium-, and small-landslide categories. These scenarios encompass a range of terrain complexities and levels of background interference. By analyzing the extraction results across different scales, the capabilities of the model in feature representation, boundary delineation, and background suppression are assessed, thereby offering further insight into robustness in complex terrain and under background interference conditions.

For large landslides, Scenario 1 [see Fig. 12(a)] features a prominent landslide with significant terrain deformation and clearly defined boundaries. The experimental results indicate that the proposed model effectively captures the spatial extent and overall morphology of the landslide, achieving high-boundary delineation accuracy. The model exhibits strong resistance to boundary blurring and feature degradation,

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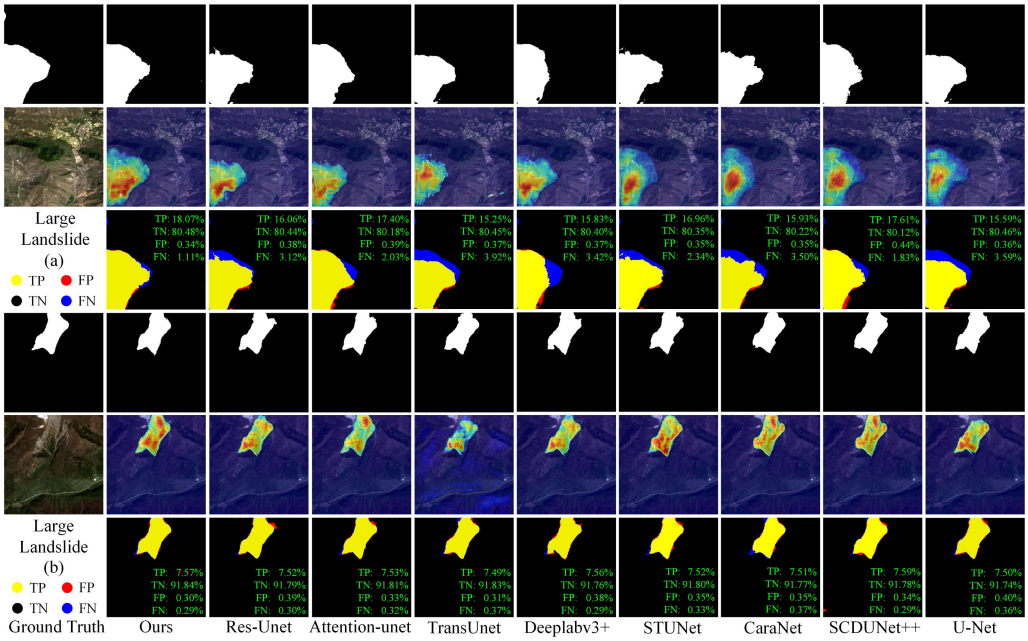



Fig. 12. (a) and (b) Visual comparison of detection results for large landslides using different models in two representative scenarios. For each scene, the first row shows the extracted landslide regions. The second row shows the feature visualization heatmaps. The third row shows the corresponding error maps, where TP, FP, TN, and FN denote the positive, false positive, true negative, and false negative, respectively.

particularly in areas with complex topographic variation and substantial background interference. Compared with Res-Unet, TransUnet, and CaraNet the proposed model achieves higher TP and TN rates, maintaining superior segmentation accuracy even under challenging conditions.

Scenario 2 [see Fig. 12(b)] depicts a large landslide situated in a relatively homogeneous terrain characterized by minimal background interference and distinct terrain transitions. In this scenario, the proposed model demonstrates high precision, effectively capturing the global shape of the landslide and preserving fine boundary details. Notably, the model suppresses background interference, reducing FPs and enhancing segmentation reliability. Compared with competing models, the proposed approach more accurately distinguishes the transition zones between landslide and nonlandslide areas, thereby preserving the structural integrity of the landslide contours.

In summary, for large-scale landslide detection, the proposed model consistently demonstrates excellent performance across both complex and simple terrain conditions. It accurately extracts salient morphological features and delineates boundaries with high precision, even in the presence of significant noise, thus highlighting its robustness, generalization ability, and stability in large landslide recognition tasks.

For medium-sized landslides, Scenario 1 [see Fig. 13(a)] presents a landslide with a relatively uniform shape but complex boundary characteristics. Medium-scale landslides often exhibit regular morphological structures coupled with intricate edge transitions, which makes them particularly vulnerable to background interference and terrain variability. The experimental results demonstrate that the proposed model effectively balances segmentation accuracy with computational efficiency in such challenging environments. It successfully captures essential morphological features and boundary details, maintaining high recognition performance despite the presence of complex background conditions. Compared with traditional models such as TransUNet and CaraNet, the proposed approach achieves higher TP and TN rates in medium-sized landslide extraction. In regions with significant topographic variation or noise interference, the model exhibits strong robustness, accurately distinguishing landslide from nonlandslide areas and significantly reducing FPs and FNs. These results underscore the resilience and reliability of the model in complex medium-scale landslide recognition tasks.

Scenario 2 [see Fig. 13(b)] features a medium-sized landslide situated in relatively smooth terrain with minimal background interference. The landslide exhibits a well-defined shape with clear boundary transitions. In this scenario, the proposed model accurately captures both the global morphological structure and detailed boundary features. Particularly at the interface between landslide and nonlandslide zones, the model effectively suppresses residual noise, thereby enhancing segmentation accuracy and overall reliability. Compared with U-Net and Attention-UNet, the proposed model demonstrates a stronger ability to preserve fine structural features, mitigating boundary blurring and local feature loss. Competing models tend to be more susceptible to terrain-induced background effects, which results in diminished boundary clarity. In contrast, the proposed model consistently maintains boundary integrity and effectively suppresses background interference, reinforcing its stability in medium-scale landslide detection.

Overall, the proposed model delivers robust and accurate performance in the identification of medium-sized landslides across diverse terrain conditions. It reliably extracts salient morphological and boundary features, demonstrating high adaptability, precision, and segmentation stability in medium-scale landslide recognition tasks.

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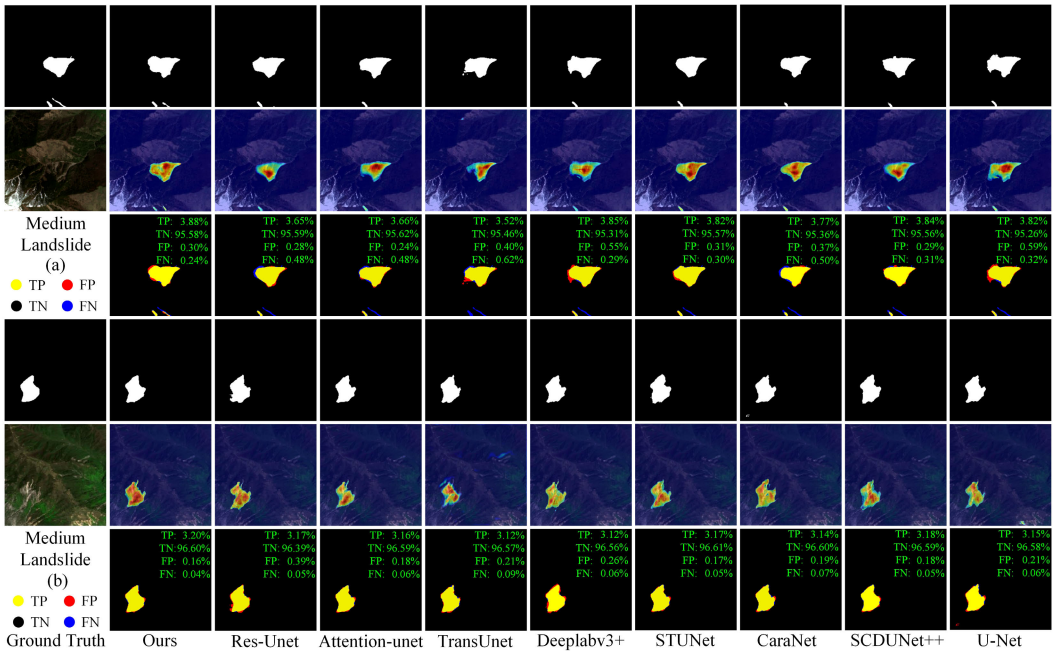



Fig. 13. (a) and (b) Visual comparison of detection results for medium landslides using different models in two representative scenarios. For each scene, the first row shows the extracted landslide regions. The second row shows the feature visualization heatmaps. The third row shows the corresponding error maps, where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative, respectively.

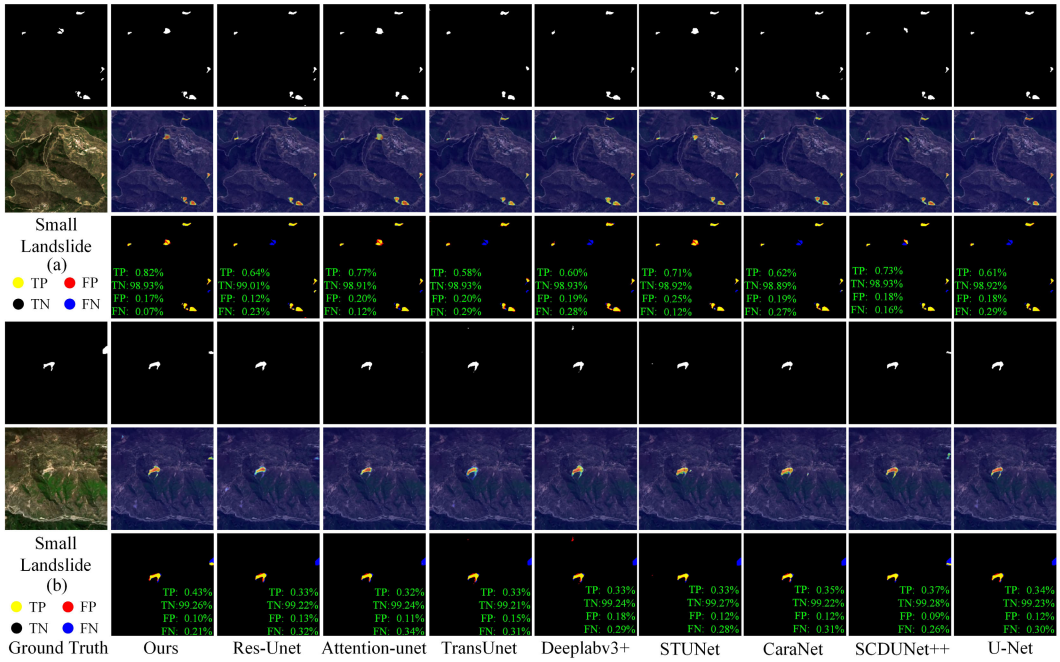



Fig. 14. (a) and (b) Visual comparison of detection results for small landslides using different models in two representative scenarios. For each scene, the first row shows the extracted landslide regions. The second row shows the feature visualization heatmaps. The third row shows the corresponding error maps, where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative, respectively.

In the context of small-scale landslide identification, Scenario 1 [see Fig. 14(a)] presents a landslide with limited spatial extent, complex terrain features, and substantial background noise. Small landslides are inherently difficult to detect because of their indistinct morphological signatures and the masking effects of surrounding topography and background interference. Experimental results indicate that the proposed model effectively captures the core characteristics of the landslide in this scenario, achieving high boundary recognition accuracy. It significantly reduces both FPs and FNs, particularly in areas with steep gradients and pronounced terrain variability. The model reliably distinguishes landslide boundaries from adjacent nonlandslide regions, thereby minimizing misclassification. Compared with existing models such

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as U-Net, CaraNet, and DeepLabv3+, the proposed model demonstrates a clear advantage in terms of the TP rate for small-scale landslide detection. This advantage is especially pronounced under complex background conditions, where the model robustly suppresses noise and enhances overall recognition performance.

Scenario 2 [see Fig. 14(b)] involves a small landslide located in a terrain context with minimal interference and relatively simple topography. The landslide is more distinct, with well-defined transitions between landslide and nonlandslide regions. Under these favorable conditions, the proposed model accurately delineates the overall shape of the landslide and exhibits superior performance in capturing fine-grained boundary details. It effectively suppresses background noise and demonstrates strong robustness in handling boundary transitions. The model achieved a marked reduction in the FP rate and demonstrated a consistent ability to differentiate between landslide and nonlandslide areas, thereby enhancing overall detection accuracy.

Across both small-scale landslide scenarios, the proposed model exhibited high segmentation precision and robustness. It reliably extracts key morphological and boundary features even in the presence of significant terrain complexity and noise. The model demonstrates excellent noise resistance and maintains high recognition performance. However, due to the inherently subtle nature of small landslides, some FNs still occur under highly complex terrain conditions, indicating room for improvement in fine-detail extraction capabilities.

In summary, the proposed model consistently achieves high accuracy, robustness, and adaptability in landslide recognition across varying spatial scales. It effectively handles diverse terrain complexities and background interference, substantially reducing FP and FN rates. These findings highlight the potential of the model as a reliable tool for landslide detection. Future research should aim to further enhance the sensitivity of the model to subtle landslide features, particularly in small-scale scenarios, by improving fine-grained feature representation and detail-level segmentation.

## V. Discussion

We first conducted a series of experiments on the proposed model using a self-constructed dataset, including ablation studies on the input multisource data and the model architecture itself. Subsequently, the generalization capability of the model is evaluated using the publicly available Landslide4Sense dataset and a self-labeled landslide dataset from the Lanzhou urban area, providing a comprehensive validation of its effectiveness.

### A. Data Ablation

To systematically evaluate the effectiveness of the proposed model under different multisource data inputs, four sets of data ablation experiments were designed: 1) only optical bands (T1); 2) optical bands + spectral indices (T2); 3) optical bands + spectral indices + terrain factors (T3); and 4) optical bands + spectral indices + terrain factors + texture indices (T4). The model performance under each input configuration was quantitatively evaluated using four metrics, precision, recall, $F1$-score, and mIoU, to comprehensively assess its robustness to varying multisource inputs. Furthermore, to validate the applicability of the model, landslides of different shapes and scales were selected for the analysis. The experimental results are shown in Fig. 15.

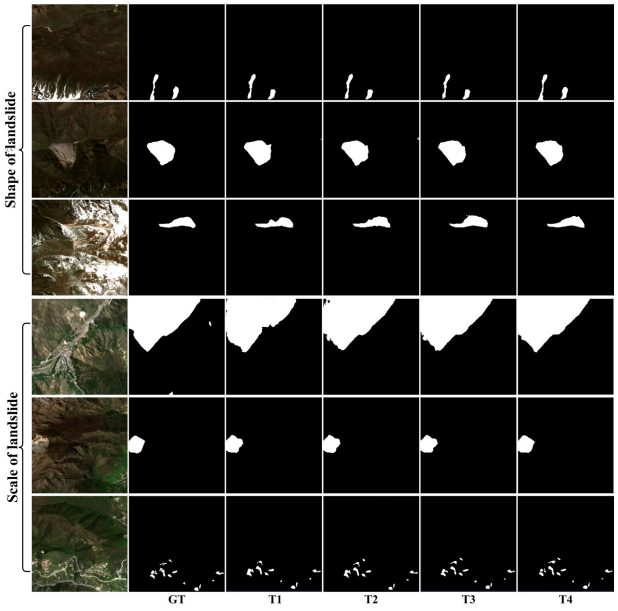



Fig. 15. Visualization results of data ablation experiments.

TABLE IV
Landslide Identification Index of Different Data Combinations

| Data | Precision(%) | Recall(%) | F1-score(%) | mIoU(%) |
|---|---|---|---|---|
| T1 | 78.80 | 70.33 | 74.57 | 75.44 |
| T2 | 76.87 | 73.64 | 75.26 | 77.62 |
| T3 | 77.42 | 74.58 | 76.00 | 78.31 |
| T4 | **80.84** | **77.89** | **79.37** | **81.74** |

The visualization results in Fig. 15 demonstrate that the proposed model can effectively identify landslides of various shapes and scales across all data input configurations, highlighting its strong generalization capability and robustness to heterogeneous data sources. In more complex scenarios, the inclusion of additional input modalities significantly enhances the ability of the model to capture critical landslide features, particularly for small-scale events and boundary regions, resulting in improved segmentation precision and the detection of subtle morphological changes. The stepwise incorporation of spectral indices, terrain factors, and texture indices progressively strengthens the feature representation capabilities of the model, addressing challenges associated with complex terrain conditions.

A comprehensive analysis of the quantitative results across the four experimental groups (T1–T4) reveals a clear trend of performance improvement with the integration of additional data sources. Compared with the baseline model utilizing only optical bands, the inclusion of spectral indices, terrain attributes, and texture information led to marked increases in precision, recall, $F1$-score, and mIoU (as shown in Table IV). In particular, the T4 configuration achieved over 80% in

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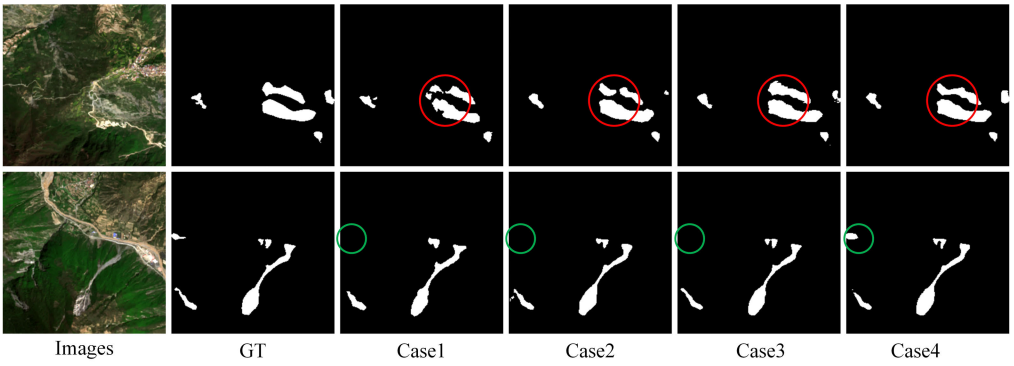
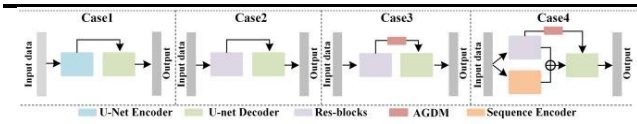



Fig. 16. Visualization results of the model ablation experiments.

both accuracy and mIoU, underscoring the effectiveness of multisource data fusion in enhancing the ability of the model to accurately delineate landslide regions. The spectral indices improved the sensitivity of the model to vegetation and surface changes, terrain factors enhanced its understanding of geomorphological context, and texture indices contributed to finer boundary delineation and the detection of small-scale landslides. In summary, the ablation experiments provide strong evidence of the generalization ability and robustness of the proposed model across diverse input configurations. The progressive integration of multisource data significantly improves landslide recognition capabilities of the model, enabling more comprehensive feature extraction and demonstrating superior adaptability to complex environmental conditions.

TABLE V
ABLATION STUDY ON MODEL COMPONENTS




| Case | Model | | | | mIoU (%) |
|---|---|---|---|---|---|
| | Baseline | Res-blocks | AGDM | MSRNet-LSTM | |
| Case1 | √ | | | | 76.72 |
| Case2 | √ | √ | | | 77.44 |
| Case3 | √ | √ | √ | | 78.13 |
| Case4 | √ | √ | √ | √ | **81.74** |

### *B. Model Ablation*

To evaluate the impact of each component of the proposed model architecture on landslide identification accuracy, four groups of ablation experiments were conducted for the landslide extraction task to analyze the contribution of each structure in landslide extraction.

*Case 1:* The baseline U-Net model.

*Case 2:* Replacing the U-Net encoder with the designed residual module.

*Case 3:* Based on Case 2, a convolutional attention mechanism (AG) is incorporated in the skip connections between the encoder and decoder.

*Case 4:* Based on Case 3, a sequential feature extraction branch (MSRNet-LSTM) is incorporated into the encoder.

The visualization results in Fig. 16 indicate that the U-Net model struggles with inaccuracies and omissions when extracting landslide boundaries in complex terrain or small-scale landslides. Incorporating residual blocks notably improves overall performance compared to the baseline U-Net, particularly in capturing the general outline and boundary details of landslide areas. The residual module enhances the compactness of feature representation by reducing feature map channels, thereby lowering computational load and strengthening the ability of the model to capture key features. However, despite these improvements, the model still exhibits limitations in accurately extracting small landslides.

Areas within highly complex scenarios are highlighted by the red circle in the figure. To address this, an attention-gating decoder module is introduced to dynamically adjust the attention of the model to different regional features. This method significantly enhances sensitivity to critical areas, especially in delineating landslide boundaries and local features. Experimental results demonstrate that the attention gating module substantially improves performance in recognizing landslide boundary regions. Compared to the first two models, the attention mechanism better captures fine details, reducing both FPs and missed detections, and enhancing the responsiveness of the model to subtle landslide features in complex scenes.

Furthermore, a sequential feature extraction branch is further integrated into the encoder to construct the complete DSCF-LDNet architecture. Experimental results demonstrate that the introduction of this branch notably enhances the model's capability to perceive cross-channel dependencies and capture fine-grained details from multisource landslide data. Consequently, the model not only achieves a more comprehensive representation of spatial features but also effectively explores the semantic relationships among heterogeneous data channels. This improvement proves particularly beneficial for the identification of small-scale and morphologically diverse landslides. Quantitative evaluations further confirm that the inclusion of this branch leads to a significant increase in mIoU (see Table V), underscoring its effectiveness in strengthening both landslide recognition performance and boundary delineation precision.

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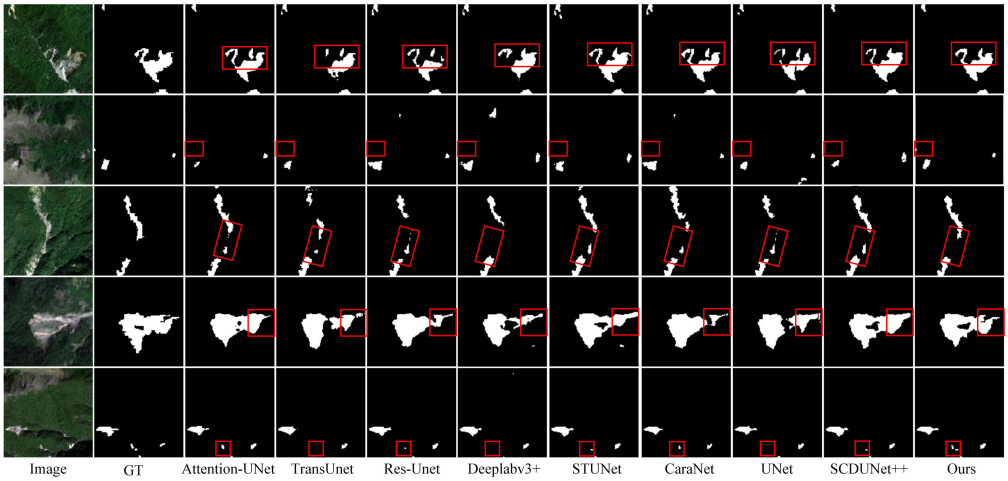



Fig. 17. Visualization of landslide detection results using different models on the Landslide4Sense dataset. The red rectangular region highlights a notable improvement in landslide target segmentation compared to other methods.

TABLE VI
EVALUATION INDEXES OF LANDSLIDE IDENTIFICATION RESULTS OF DIFFERENT MODELS

| Model | Precision(%) | Recall(%) | F1-score(%) | mIoU(%) | Params(M) | FLOPs(G) |
|---|---|---|---|---|---|---|
| U-Net | 79.23 | 63.57 | 70.54 | 70.19 | 30.16 | 147.65 |
| ResU-Net | 77.15 | 70.32 | 73.54 | 72.34 | 43.93 | **84.69** |
| Attention-UNet | 78.66 | 71.69 | 75.01 | 73.48 | **33.18** | 189.51 |
| TransUNet | 73.26 | 64.02 | 68.33 | 68.54 | 105.34 | 93.24 |
| Deeplabv3+ | 71.74 | 66.78 | 69.16 | 69.32 | 68.70 | 192.87 |
| STUNet | 78.21 | 71.32 | 74.77 | 73.61 | 41.64 | 135.42 |
| SCDUNet++ | 79.76 | 71.51 | 75.63 | 74.46 | 63.20 | 118.49 |
| CaraNet | 77.26 | 71.08 | 74.04 | 71.83 | 47.59 | 121.37 |
| Ours | **81.02** | **73.92** | **77.30** | **76.27** | 54.72 | 102.41 |

### C. Model Application

*1) Application of the Landslide4Sense Dataset:* We further evaluated the performance of DSCF-LDnet using the public Landslide4Sense dataset, which is a benchmark for landslide remote sensing detection [31]. The Landslide4Sense dataset consisted of real landslide images collected from four different geographical locations and times: Iburi (September 2018), Kodagu (August 2018), Gorkha (April 2015), and Taiwan (August 2009). The dataset contains 3799 remote sensing images, each of which is a fusion of multispectral optical images (B1–B12) captured by the Sentinel-2 sensor, along with DEM and slope data from the ALOS phased-array L-band SAR (PALSAR). Specifically, it can be abstracted as a 14-channel image for processing.

Before training the DSCF-LDnet model, data enhancement was applied to the Landslide4Sense samples to improve diversity. The specific methods are detailed in Section III-C. After augmentation, the samples were randomly split into training, validation, and testing sets in a 7:2:1 ratio. During training, the input images were upsampled to a 256 × 256 resolution, with the batch size set to 4 and the initial learning rate configured as $\lambda = 10^{-5}$, and chose the Adam optimizer to adaptively adjust the learning rate size. In addition, all experiments were performed under the same hardware and software conditions described in Section III-C.

After training, the testing set is used to evaluate the effectiveness of the learned weights. For a comprehensive assessment, we also conducted a comparative analysis on the Landslide4Sense dataset using several mainstream models introduced in Section IV-A. The visual results of the landslide extraction and segmentation are presented in Fig. 17. It can be seen that DSCF-LDnet successfully extracts landslide targets of various sizes and shapes from the Landslide4Sense dataset, outperforming other methods in distinguishing landslide targets from visually similar areas, such as bare land. The red rectangular region in Fig. 17 highlights a significant improvement in landslide target segmentation compared to alternative approaches. Performance comparisons are summarized in Table VI, where the proposed DSCF-LDnet model achieved superior results at comparable computational costs. Specifically, it achieved the highest accuracy of 81.02%, the recall rate of 73.92%, the $F1$-score of 77.30%, and the mIoU of 76.27%. These results demonstrate that the proposed method effectively isolates landslide targets from complex remote sensing imagery. When using multisource data inputs, DSCF-LDnet exhibits clear advantages over other DL segmentation models, underscoring its suitability for practical applications.

*2) Application of Lanzhou Urban Datasets:* To evaluate the adaptability of the proposed approach across different scenarios, we tested the DSCF-LDnet model using the Lanzhou

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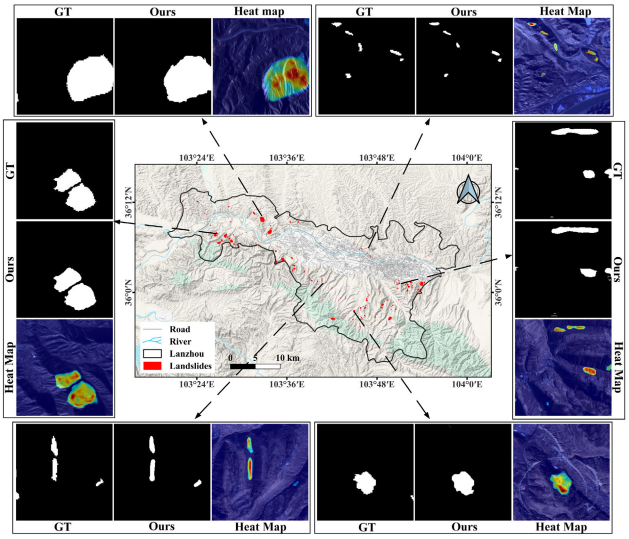



Fig. 18. Visualization of landslide detection results in Lanzhou urban area using the proposed mode.

urban landslide dataset, as used in the literature [32]. In this experiment, the dataset was divided into training, validation, and testing sets at a ratio of 7:2:1. The model training strategy, initial hyperparameter settings, and software and hardware environment were kept consistent with the Bailong River basin landslide experiment. The experimental results demonstrate that the DSCF-LDnet model performs exceptionally well on Lanzhou urban landslide dataset, achieving an accuracy rate of 86.42%, recall rate of 82.81%, $F1$-score of 84.62%, and mIoU of 83.79%. These results further confirm that the DSCF-LDnet model possesses robust feature learning capabilities across different experimental conditions, enabling high-precision landslide recognition in remote sensing images with complex backgrounds. In addition, some of the landslide detection results are shown in Fig. 18, where each rectangular box in the figure represents real landslide samples, the prediction results of the proposed model, and visual thermal maps of features from left to right. The results indicate that DSCF-LDnet can achieve accurate landslide identification on the Lanzhou urban dataset.

In summary, the experimental results based on the Landslide4Sense dataset and the Lanzhou urban landslide dataset fully verify the superior performance of the proposed DSCF-LDnet model under different geographical regions, data types, and terrain conditions. Whether applied to a public benchmark dataset with multisource heterogeneous data fusion or to landslide datasets with complex backgrounds, the model consistently maintains high recognition accuracy and stable boundary segmentation performance, successfully extracting landslide targets of varying scales and morphologies. The results indicate that DSCF-LDnet effectively adapts to feature differences across diverse scenarios, demonstrating strong feature learning and generalization capabilities.

## VI. Conclusion

This study proposes a DSCF-LDNet. The model employs a dual-stream encoder, with a spatial branch extracting fine-grained spatial features and a sequential branch capturing interchannel semantic dependencies of fused multisource features. The two types of features are integrated through a feature fusion module, and an attention gating decoder module in the decoding stage further enhances the model's discriminative capability for landslide regions.

Experimental results demonstrate that DSCF-LDNet achieves high accuracy and robustness in feature extraction and boundary delineation for landslides of various shapes and scales. In complex terrain and high-background-noise conditions, the model effectively suppresses interfering signals, significantly reducing FPs and FNs. On the self-constructed Bailong River Basin dataset, DSCF-LDNet attains a precision of 80.84%, a recall of 77.89%, an $F1$-score of 79.35%, and an mIoU of 81.74%, outperforming SOTA models including U-Net, ResU-Net, Attention U-Net, DeepLabV3+, STUNet, CaraNet, and SCDUNet++. Validation on the Landslide4Sense and Lanzhou urban landslide datasets further confirms the model's generalization and robustness across diverse geographic and geological conditions. Overall, DSCF-LDNet provides an effective solution for high-precision landslide detection in complex geophysical environments.

The current evaluation of the model's generalization relies solely on the Landslide4Sense and Lanzhou urban landslide datasets, which have limited sample types and cannot fully cover landslides of different types and triggering conditions. Existing experiments mainly assess the model's generalization using train-test splits within the same dataset. Future work will conduct cross-regional experiments, such as training in one region and testing in areas with significantly different geomorphological or ecological conditions, and will explore strategies like transfer learning or domain adaptation to enhance the model's adaptability and generalization across heterogeneous regions. In addition, lightweight versions of DSCF-LDNet will be developed to support real-time monitoring applications.

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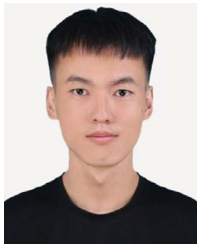
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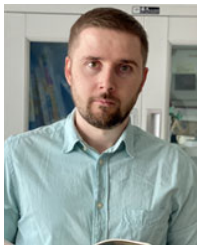
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